

ASSESSMENT BRIEF (COURSEWORK 2)

Programme	BSc Computer Science/ BEng Software Engineering		
Module Title	Data Engineering		
Module Code	CMP-X304-0		
Module Level	6	Assessment Type(s)	Report and Presentation (both mandatory to pass)
Word Length / Duration	500 words	% contribution to module mark	50%
Deadline (date & time) for Submission	24 th April 2026	Format/Location of submission	PDF/ Moodle
Assessment Feedback date:	3 weeks after the demo		
Learning Outcomes This assessment has been designed to provide you with an opportunity to demonstrate your achievement of the learning outcomes listed below. By successfully completing this assessment, you will be able... LO5: Analyse the requirements for building a data delivery pipeline. LO6: Work as a member of a development team recognising the different roles within a team and different ways of organising teams.		Employability and Professional Skills 1. Cloud & Data Engineering Skills Development Students gain hands-on experience with industry-standard tools such as Azure Data Factory, Databricks, Data Lake Gen2, and Synapse Analytics, building practical skills highly valued in data engineering and cloud computing roles. 2. End-to-End Data Pipeline Experience The coursework mirrors real-world data engineering workflows—from ingestion and transformation to storage and visualization, enhancing students' ability to design scalable, production-ready data solutions. 3. Teamwork and Professional Collaboration As a group project, it develops collaboration, communication, and role-based working skills, reflecting	

	professional team structures in tech organisations. 4. Presentation & Communication Skills Through report writing, video demonstrations, and PowerPoint presentations, students build the ability to clearly communicate technical concepts, an essential skill for client-facing and stakeholder-driven roles.
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Assessment Requirements

The final mark will come from two components (Both components are mandatory to pass this assignment)

1. Submission through Moodle.
2. Presentation.

The report consisting of the video links (**for tasks 1, 2, 4 and 5**) and written content (**for task 3 and 4**) must be submitted within the due date to be considered for a complete submission for this assessment. This assessment is **GROUP-BASED** and **only one member of the group will upload the document**. Presentation will only be held after the work has been submitted through the specified link on Moodle. Groups will be the same as Coursework 1 and **all members must contribute** to the completion of all tasks. Task allocation and internal coordination are the responsibility of the group.

Coursework Description:

Objective: To create a data pipeline in Azure Cloud that ingests data from a given source, loads it into Azure Data Lake Gen2, transforms it, loads it back into Azure Data Lake Gen2 and finally do some visualization in Azure Synapse Analytics.

The coursework is worth 50% of the total module mark. It is a group project in which the students will:

Setup Azure Account: If you do not have an Azure account, sign up for a free student account.

Create Azure Data Factory: Navigate to Azure Portal and create a new Data Factory resource and then configure linked services for your data sources and destinations and ingest data using Data Factory.

Data Transformation: Use Azure Databricks to transform the data as specified in the tasks below.

Data Loading: Load the transformed data into Azure Data Lake Gen2.

Data Visualization: Visualize the data in Azure Synapse Analytics.

I. Task Description

Task 1. Data Ingestion

- Ingest the 6 files provided at <https://github.com/SameenaNaaz/Datasette> to the Azure Data Lake Gen 2.

Task 2. Data Processing

- Use Azure Databricks to process and transform the data as required.
- Apply data cleaning, filtering, aggregations, or any other necessary transformations.

Task 3. Data Storage

- Store the processed data in Azure Data Lake Gen2.

Task 4: Data Visualization

- Visualize the processed data using Azure Synapse Analytics.

Tasks	Activities	Submission
Task 1	• Create Azure Storage Account • Create Azure Data Factory and create two directories raw_data and transformed_data for storing the raw data and the transformed data respectively. • Ingest the 6 csv files from https://github.com/SameenaNaaz/Datasette into the raw_data directory.	Include a hyperlink in your PDF report that directs to your video stored on OneDrive Explain the steps performed for carrying out the task till ingestion of any one csv file in your video
Task 2	• Create an instance of Databricks and mount ADLS Gen2 in Azure Databricks. • Read all the files from the raw_data directory in ADLS Gen2	Include a hyperlink in your PDF report that directs to your video stored on OneDrive This video should explain the deployment of Azure Databricks and creation of compute resource.
Task 3	• Apply the following transformations in Azure Databricks and write back the transformed files in the transformed_data directory in ADLS Gen2. ◆ Retain back only the first 12 columns from executives.csv. ◆ Retain back only the first 12 columns from legislator_terms.csv and remove any rows with missing values.	In your report, discuss about your code for mounting of ADLS Gen2 in Azure Databricks and for carrying out the transformations and the rationale behind it within 200 -250 words) for Task 3. You must provide the screenshots of each operation/execution of the code created by you with proper commenting.

	◆ Write a sql spark query to count the unique number of Male and Female legislators from the legislators.csv file. ◆ Create two new columns "start_year" and "end_year" extracting the year from the "start" and "end" columns in the executive_terms.csv file.	
Task 4	• Create an instance of Synapse Analytics and copy the csv files from the transformed_data directory into the workspace. • Apply different sql queries, one on each of the csv files and show the outcome in the form of chart.	Include a hyperlink in your PDF report that directs to your video stored on OneDrive. The video should show the steps for creating an instance of Synapse Analytics and copying any one csv file. In your report, discuss about the sql queries executed for achieving task - 4 (within 200 - 250 words). You must provide the screenshots of the queries along with their output both in tabular and graphical form.
Task 5	• Use Power BI in Synapse analytics and create a dashboard for visualization purpose.	Include a hyperlink in your PDF report that directs to your video stored on OneDrive showing the steps performed

II. Submission Requirements:

Report in PDF format

The report should use 12-point font in Times New Roman, 1-inch margins, and double line spaced. The report should be properly paged, paragraphed, and sectioned, and include the following sections in order in a (.pdf) file.

1. Cover Page with names of all the members of the group.
2. Video link/ written description as per the specification in the above table

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed.

Include a hyperlink in your PDF report that directs to your video (**Tasks 1, 2, 4 and 5**) stored on OneDrive and ensure that appropriate access permissions are granted; otherwise, the submission will not be marked. Each member of the group must demonstrate some tasks in the video and the recording should have the picture of the presenter along with the voiceover.

III. Marking Scheme

Task 1: 15 marks

Task 2: 15 marks

Task 3: 15 marks

Task 4: 15 marks

Task 5: 10 marks

Presentation (Dates to be confirmed soon): 30 marks.

Each group must prepare a PowerPoint presentation of maximum 10 slides.

Marking Criteria

Task / Criteria	Outstanding	Excellent	Good	Fair	Poor	Fail (0 marks)
Task 1: Data Ingestion	Data successfully ingested into ADLS Gen2 with fully configured Azure Data Factory. Clear structure and strong justification.	Correct ingestion with minor issues. Good structure and explanation.	Functional ingestion with acceptable results but limited optimisation.	Basic ingestion with inconsistencies or partial completion.	Attempt made but major issues affecting success.	No submission or non-functional ingestion.
	15 marks	12 marks	10 marks	7 marks	4 marks	0 marks
Task 2: Data Processing (Databricks)	Efficient and well-structured transformations. Optimised and clearly justified.	Correct transformations with minor inefficiencies. Good explanation.	Functional transformations with limited optimisation.	Basic transformations with errors or incomplete logic.	Attempted but incorrect or poorly executed.	No meaningful attempt.
	15 marks	12 marks	10 marks	7 marks	4 marks	0 marks
Task 3: Data Storage & Transformation	All transformations accurately implemented and stored correctly. Strong justification.	Most transformations correct with minor issues.	Adequate implementation with some missing elements.	Partial implementation with noticeable errors.	Major issues in transformation or storage.	No submission or incorrect implementation.
	15 marks	12 marks	10 marks	7 marks	4 marks	0 marks
Task 4: Data Visualization (Synapse)	Advanced visualisations with correct SQL queries and strong analytical insight.	Good visualisations with correct queries and reasonable interpretation.	Basic visualisations with limited insight.	Minimal visualisation with weak or incorrect queries.	Attempted but poor outputs.	No visualisation.
	15 marks	12 marks	10 marks	7 marks	4 marks	0 marks
Task 5: Power BI Dashboard	Highly interactive and insightful	Functional dashboard	Basic dashboard	Minimal dashboard	Attempted but	No dashboard created.

	dashboard with professional design.	with good visuals.	with limited features.	with poor clarity.	ineffective dashboard.	
	10 marks	8 marks	6 marks	4 marks	2 marks	0 marks
Presentation	Comprehensive understanding, confident delivery, excellent responses.	Strong understanding with clear explanations.	Reasonable understanding, answers basic questions.	Limited understanding, struggles with clarity.	Very weak understanding and poor communication.	No presentation or engagement.
Marks	30 marks	25 marks	20 marks	15 marks	9 marks	0 marks

Assessment Success Guidance

To achieve high marks, students should demonstrate more than basic task completion. Strong submissions will show:

- Robust Azure pipeline implementation – A fully functional end-to-end pipeline (Data Factory → Data Lake → Databricks → Synapse → Power BI) with accurate and well-structured outputs.
- Effective use of cloud tools – Correct configuration and integration of Azure services demonstrating industry-relevant skills.
- Clear justification of decisions – Explanation of transformations, storage choices, and overall pipeline design.
- High-quality data processing and visualisation – Accurate transformations and meaningful insights presented through SQL queries and dashboards.
- Professional presentation of work – Well-structured report, clear screenshots, and properly commented code.
- Teamwork and confident presentation – Evidence of collaboration, clear individual contribution, and ability to explain work effectively.

Use of Artificial Intelligence (AI)

Please read and apply:

<https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4a%20AI%20principles%20and%20guidance%20-for%20Senate.docx>

https://roehamptonprod.sharepoint.com/sites/portal/information/LTEU/Documents/4b%20Student%20Guidelines%20on%20the%20use%20of%20AI_.docx

The assessment is designed so that the use of AI during the assessment is possible. You must acknowledge any use of AI and appropriately cite all AI generated outputs. Please make sure you read and understand the assessment guidelines and ask your Module Leader if you have any questions.

**You can find the student guidance on the use of AI in the Library and Nest*

Referencing

References: Referencing any point in your discussion section(s). If you have used ideas from anywhere other than the lecture notes and tutorial examples (e.g., from a book, the Internet, or a fellow student) then include a reference showing where the code or ideas came from and label your code carefully to show which parts are your and which parts are borrowed. You should include your reference list at the end of your report. You must ensure you use the appropriate IEEE format for your subject area. Your referencing must use the IEEE referencing style IEEE Citation Guidelines. It is highly recommended that you use reference management software such as RefWorks that is provided by the university. Your report should have as many references as is required. It is your responsibility to ensure that you have actually read all the material you reference, and that the references provided in your report are legitimate and NOT AI-generated.
Mitigating circumstances/late penalties Sometimes circumstances outside of your control may affect your studies and might prevent you from submitting work on time or attending an exam. The University offers the ability for students to request additional time to complete an assessment or to defer an examination to a later date. If you are finding yourself in such a situation, please speak to your Academic Guidance Tutor, the Roehampton Student Union (RSU) or someone in the <u>Wellbeing team</u> first, who can support you. Further details can be found on the mitigating circumstances portal. If you do not apply for or are not approved for Mitigating Circumstances, late penalties will apply. If work is submitted up to 14 days late, the mark will be capped at 40%; if it is over 14 days late, it will not be marked.
Resubmissions and Reassessment If you are required to resubmit this assessment or take part in reassessment, you will be notified via Moodle and your student email. Please ensure you check both regularly. Any reassessment tasks will follow the same learning outcomes and criteria.
Submission Checklist Before you submit, ask yourself: Have I fully answered the assessment brief? Have I met the word count and formatting requirements? Is my referencing complete and accurate? Have I declared any AI use honestly? Have I proofread my work? Am I submitting through the correct platform before the deadline?